



Syllabus

Introduction

This course investigates the fundamental issues addressed by scientists studying tropical rain forests, including what gave rise to their remarkable biodiversity, the drivers and consequences of deforestation, why people are fascinated by rain forests, cultural and ecological stereotypes about the tropics, and different approaches to forest conservation and sustainability. ***By the end of the course students will be able to:***

- Recognize and describe stereotypes about rain forests & their residents
- Analyze rain forest tropes in art, literature, & popular culture
- Discuss & evaluate hypotheses for the origins and maintenance of tropical biodiversity
- Explain & compare the history and ecological influence of humans in rain forests
- Review contemporary threats to rain forests
- Analyze and visualize data on deforestation
- Review and contrast strategies for rain forest conservation & restoration
- Identify the presence of rain forests in their daily lives & set personal goals for advancing rain forest conservation
- Produce materials for communicating about tropical biology & the conservation of tropical rain forests to family and peers

When & Where

When: Tuesdays 11:45 AM-12:35 PM & Thursdays 11:45 AM-1:40 PM

Where: Larsen Hall 0310

Instructor & TA

Instructor: Dr. Emilio M. Bruna, embruna@ufl.edu, (352) 846-0634)

Teaching Assistant: Ashra Kunwar, kunwarashra@ufl.edu

Credits & Prerequisites

Credits: 3

Prerequisites: None

Quest Designation: Quest 2

GenEd Designation: International

- A minimum grade of C is required for Quest and General Education credit.
- Courses intended to satisfy Quest and GenEd requirements *cannot* be taken S-U.

Majors, Minors, and Certificates

This course counts towards the graduation requirements of the following UF Majors, Minors, and Certificates:

- **Majors:**
 - Sustainability Studies ([info](#))
 - International Studies ([info](#))
- **Minors:**
 - Latin American Studies ([info](#))
 - Sustainability Studies ([info](#))

Required Course Materials

Materials and Supplies Fees: None.

Textbooks to purchase: None. All materials, including readings and videos, will be made available on the course Canvas page. *Students must sign up for free online access to the New York Times, Wall Street Journal, and The Economist by following the instructions at [this UF Libraries Website: https://businesslibrary.uflib.ufl.edu/ws-j-nyt-economist](https://businesslibrary.uflib.ufl.edu/ws-j-nyt-economist).*

Office Hours

💡 Can you give me **one good reason** why I should go to Office Hours?

I can give you **ten**.

1. To introduce yourself.
2. Get clarification on assignments.
3. Argue about a topic that came up in class.
4. Grab a (free) tea, coffee, or espresso in our lab kitchen.
5. Make sure you understood the key points from a class session.
6. Ask for feedback on ideas for course projects.
7. Get advice on successfully navigating academic life at UF.
8. Discuss how to gain experience for your post-graduation goals.
9. Request help arranging a study group.
10. You don't need a good reason...just come on by.

Instructor

- **When:** Wednesday 1:30-3:30 or by appointment. To guarantee a specific time slot you can sign up here: <https://embruna.youcanbook.me/>. If you can't make it during the regularly scheduled office hours, you can send a message via Canvas to set up an appointment.
- **In-person:** The Tropical Ecology & Conservation Lab is located at 711 Newell Drive next to Rawlings Hall (between the bus stop and the parking garage). The location and directions are on [Google Maps](#).
- **Online:** To join Office Hours virtually use the Zoom Conferences link on the main menu of the course Canvas page. We are online the entire session.

Teaching Assistant

When: Tuesday 4:00-5:30 pm or by appointment.

Assignments & Grades

Learning in our class is achieved with an diverse array of methods ranging from data analysis to essays to projects. In most class sessions you will also be working with small groups of students to complete an in-class assignment that reinforces the major themes of the day's topic. In keeping with the philosophy of the Quest program, this course also has *Experiential Learning and Self-Reflection Components*.

 Important note regarding class discussions & group work.

We will explore some challenging, important problems and increase our understandings of different perspectives and approaches for addressing them. These conversations may not always be easy; we sometimes will make mistakes in both how we communicate our perspective and what we hear other say. There may be times when we need patience, courage, imagination, and of course mutual respect to engage our texts, classmates, instructors, guests, and our own ideas and experiences. *Disrespectful or disruptive behavior will not be tolerated*. And always remember that as scholars we rely on critical thinking, data, prior scholarship, and expert opinion when interrogating assigned readings and discussing course content with classmates and instructors.

Assignments (1000 pts total)

- 1) In-class activities: 550 (55%)
- 2) Analytic Essay: 200 (20%)
- 3) Reflective Essay: 150 (15%)
- 4) Movie Reviews: 100 (10%)

Grading

Due dates: Due dates of Essays and Movie Reviews are posted on the course calendar. **Submissions of in-class assignments are due by the following class session.** Late assignments will lose 10 pts.

Regrades: Requests for re-evaluation of assignments must be accompanied by an explanation for why you think you deserve additional credit and the number of additional points you think you deserve. The deadline for submission is one week after the work was returned.

Grade Assignment (based on % of total points earned): A = 94–100%, A- = 90–93%, B+ = 87–89%, B = 84–86%, B- = 80–83%, C+ = 77–79%, C = 74–76%, C- = 70–73%, D+ = 67–69%, D = 64–66%, D- = 60–63%, E < 60

Grade Points: For information on how UF assigns grade points, visit: <https://catalog.ufl.edu/UGRD/academic-regulations/grades-grading-policies/>

Attendance

Though attendance is not required, many of the sessions we will be completing activities in class that count towards your grade. Some of these can be completed independently outside of the class session, but by doing them in class you will benefit from working collaboratively with the other students. **Not all in-class activities can be completed independently - if you aren't in class that day you won't be able to complete the assignment.** However, there are opportunities for extra credit to offset this, and only a subset of assignments will count towards the final class grade. ***If you need to miss class for any reason, please let me know as soon as possible.*** We will try to make arrangements for you to complete any assignments and go over any material you will be missing.

Participation

Consistent informed, thoughtful, and considerate class participation is encouraged (and in some cases required). *If you have personal issues that prohibit you from joining freely in class discussion (e.g., shyness, language barriers, medical condition): no problem.* let us know and we will discuss alternative modes of participation.

Course Calendar

Course Topics and Key Dates

WEEK	DATE	TOPIC	READ	KEY DATE
Wk 1	20-Aug	Course Welcome and Introduction		
WHY ARE WE FASCINATED BY TROPICAL RAIN FORESTS?				

Wk 2	25-Aug	Historical Narratives	1	
	27-Aug	Historical Narratives, cont.		
Wk 3	1-Sep	Rain forests in Art & Literature	2	
	3-Sep	Rain forests in Pop Culture	3	Movie Reviews Assigned
THE ECOLOGY & EVOLUTION OF TROPICAL RAIN FORESTS				
Wk 4	8-Sep	What is a rain forest?	4,5	
	10-Sep	Where is a rain forest?		
Wk 5	15-Sep	Patterns of biodiversity	6,7	
	17-Sep	Patterns of biodiversity, cont.	8	
Wk 6	22-Sep	Origins of tropical biodiversity		
	24-Sep	Maintenance of biodiversity		
Wk 7	29-Sep	Humans in rain forests	9,10	
	1-Oct	Humans in rain forests, cont.		Movie Reviews Due
Wk 8	6-Oct	The 'Paradox of Luxuriance'	11,12	
	8-Oct	Rain forest generalizations (revisited)		JUNGLE FILM FESTIVAL
THE DRIVERS AND IMPACTS OF DEFORESTATION				
Wk 9	13-Oct	How much rain forest is there?	13	Analytic Essay Assigned
	15-Oct	How much have we lost?	14	
Wk 10	20-Oct	Drivers of deforestation	15-17	
	22-Oct	Drivers of deforestation, cont.	18,19	
Wk 11	27-Oct	Rain forests & climate	20,21	
	29-Oct	Rain forests & climate change	22,23	Analytic Essay Due
THE FUTURE OF TROPICAL RAIN FORESTS				
Wk 12	3-Nov	International conservation frameworks	24-26	Reflective Essay Assigned
	5-Nov	International trade in tropical commodities	27-30	DURIAN FEST
Wk 13	10-Nov	Consumer choices & rain forest conservation	31-35	
	12-Nov	Community-based conservation	36-39	Reflective Essay Due
Wk 14	17-Nov	Thanksgiving Break		
	19-Nov	Thanksgiving Break		
Wk 15	24-Nov	Protected Areas	40-42	
	26-Nov	Reforestation & Regeneration	43-45	
Wk 16	1-Dec	Rain forests and global health	46-48	
Finals	10-Dec	No Final Exam		

Frequently Asked Questions

- (1) **How do I contact the Instructor or TA?** Send a message in CANVAS. That way both the TA and instructor will see it and you will get a response more quickly (emails sent to us directly tend to get buried under other messages).
- (2) **How will you send class announcements?** CANVAS. Check the course page for announcements and be sure you are receiving Canvas emails and updates.
- (3) **What work should we complete BEFORE class?** Read, watch, listen to, or review all materials assigned for the session. This material sets the stage for in-class activities.

- (4) **What will we do DURING class?** In-class exercises that reinforce key concepts, discussions of the assigned readings, and let you practice skills in other assignments. Some are completed individually, while others require working in groups or pairs. Each activity will have instructions and a rubric; most are designed to be finished in class.
- (5) **When is the “in-class” work due?** Deadlines are in Canvas, but typically the start of the next class session.
- (6) **What if I miss class?** Attendance is not required, but in many of the sessions we will be completing activities in class that count towards your grade. Some of these can be completed independently, but by doing them in class you will benefit from working collaboratively with the other students. However, some of the in-class activities can not be completed on your own. That’s why only a subset of the in-class assignments count towards your grade and we offer extra credit.
- (7) **I have to miss class on a certain date. What should I do?** Let us know as soon as possible so we can make arrangements for you to review material you will miss and complete assignments.”)
- (8) **Class discussions are difficult for me. Will this affect my grade?** No! If there are issues that make engaging in discussions difficult (e.g., shyness, language barriers, a medical condition), let us know and we will find alternative modes of participation.
- (9) **I unexpectedly have no child care today...what can I do?** UF does not have a policy on children in the classroom, so here is mine: I never want students to feel they have to choose between their responsibilities as parents and their education. Bringing your child to class because of a gap in child care is totally fine (and that includes babies). If you do so, please try and sit close to the exit so that you can more easily step outside if you need to care for them and so other students can continue learning.

UF Policies and Resources

Complete information about grading policies, support for students with disabilities, course evaluations, the Honor Code, Software Use, and other course policies and campus resources can be found on the [Syllabus Policies page](#). Information about university attendance policies, including policies regarding religious holidays, illness, and the twelve-day rule, can be found in the [Undergraduate Catalog](#).

In-Class Recording

Students are allowed to record video or audio of class lectures. However, the purposes for which these recordings may be used are strictly controlled. The only allowable purposes are: (1) for personal educational use, (2) in connection with a complaint to the university, or (3) as evidence in, or in preparation for, a criminal or civil proceeding. All other purposes are prohibited. Specifically, students may not publish recorded lectures without the written consent of the instructor. **A ‘class lecture’ is:** an educational presentation intended to inform or teach enrolled students about a particular subject, including any instructor-led discussions that form part of the presentation, and delivered by any instructor hired or appointed by the University, or by a guest instructor, as part of a UF course. **A**

class lecture does not include: lab sessions, student presentations, clinical presentations such as patient history, academic exercises involving solely student participation, assessments (quizzes, tests, exams), field trips, private conversations between students in the class or between a student and the faculty or lecturer during a class session. **Publication without permission of the instructor is prohibited. To ‘publish’ means:** to share, transmit, circulate, distribute, or provide access to a recording, regardless of format or medium, to another person (or persons), including but not limited to another student within the same class section. Additionally, a recording, or transcript of a recording, is considered published if it is posted on or uploaded to, in whole or in part, any media platform, including but not limited to social media, book, magazine, newspaper, leaflet, or third-party note/tutoring services. ***A student who publishes a recording without written consent may be subject to a civil cause of action instituted by a person injured by the publication and/or discipline under UF Regulation 4.040 Student Honor Code and Student Conduct Code.***

Student Learning Objectives

Content

Students demonstrate competence in the terminology, concepts, theories, and methodologies used within the discipline(s). (B = Biological Sciences, Q2 = Quest 2)

- (1) Describe the evolutionary and ecological factors underlying the distribution of biodiversity in tropical rain forests. *Assessments:* In-class activities. (B).
- (2) Distinguish (a) the different ways humans use and alter rain forests, and (b) the social, economic, and biological consequences of these activities. *Assessments:* In-class activities, Reflective Essay. (B, Q2)
- (3) Explain how genetics, remote sensing, computational tools, and other scientific developments have advanced research on the ecology and evolution of rain forest biota. *Assessments:* In-class activities, Analytic Essay. (B).
- (4) Examine the historical factors that have shaped cultural perceptions of rain forests. *Assessments:* In-class activities, Movie Reviews. (Q2).
- (5) Analyze how narratives about rain forests are reflected in different types of contemporary cultural expression. *Assessments:* In-class activities, Movie Reviews. (Q2).
- (6) Distinguish between the primary drivers of forest loss and how they vary geographically. *Assessments:* In-class activities, Analytic Essay. (B, Q2)
- (7) Critique different proposed mechanisms for rain forest conservation. *Assessments:* In-class activities, Analytic Essay. (B, Q2).

Critical Thinking

Students carefully and logically analyze information from multiple perspectives and develop reasoned solutions to problems within the discipline(s). (B = Biological Sciences, Q2 = Quest 2)

- (1) Test hypotheses regarding trends in deforestation and how forest regeneration varies geographically with quantitative data on forest cover and use news stories, primary literature, and other sources to propose mechanisms responsible for the patterns observed. *Assessments:* In-class activities, Analytic Essay (B, Q2).
- (2) Analyze qualitative data on rural-urban migration and human demographic shifts in tropical countries and assess the potential implications of the results for conservation and broader societal issues. *Assessments:* In-class activities, Analytic Essay (B, Q2).
- (3) Explain the relationship between deforestation and the global climate cycle. Compare alternative policy pathways for reducing CO₂ emissions using the En-ROADS global climate simulator. *Assessments:* In-class activities, Reflective Essay (B, Q2).

Communication

Students communicate knowledge, ideas and reasoning clearly and effectively in written and oral forms appropriate to the discipline(s). (B = Biological Sciences, Q2 = Quest 2)

- (1) Generate graphical analyses of quantitative data used to study rain forest biology and conservation (Q2, B). *Assessments:* In-class activities, Analytic Essay.
- (2) Compose summaries of research on multidisciplinary questions relevant to tropical forest biology and conservation based on logical arguments (Q2, B). *Assessments:* In-class activities, Analytic Essay.
- (3) Develop audience-specific content and materials with which to communicate an important issue related to tropical forest biology and conservation. *Assessments:* In-class activities, Reflective Essay, Movie Reviews, Analytic Essay, Final Project (Q2).

Connection

Students connect course content with meaningful critical reflection on their intellectual, personal, and professional development at UF and beyond. (B = Biological Sciences, Q2 = Quest 2)

- (1) Examine the cultural, economic, and historical experiences of people in rain forest countries, how this compares with our preconceived notions of the same, and the consequences of this disparity for our scientific understanding of rain forests and their conservation. *Assessments:* In-class activities, Movie Reviews, Final Project (Q2).

- (2) Assess the feedbacks between (a) the actions of individuals, governments, and the private sector, (b) global economic, social, and political conditions, and (c) the status rain forests and the global climate cycle. *Assessments:* In-class activities, Reflective Essay, Analytic Essay (Q2).
- (3) Appraise the ubiquity of tropical forest products in daily life, the role and impact of the global commodity chains that make this possible, and the consequences of consumer behavior for forest conservation and socioeconomic sustainability. *Assessments:* In-class activities, Final Project (Q2).
- (4) Analyze the local, regional, and global ecosystem services provided by tropical rain forests, how these vary geographically, and some of the cultural and ecological factors responsible for these differences. *Assessments:* In-class activities, Reflective Essay, Analytic Essay, Final Project (Q2)